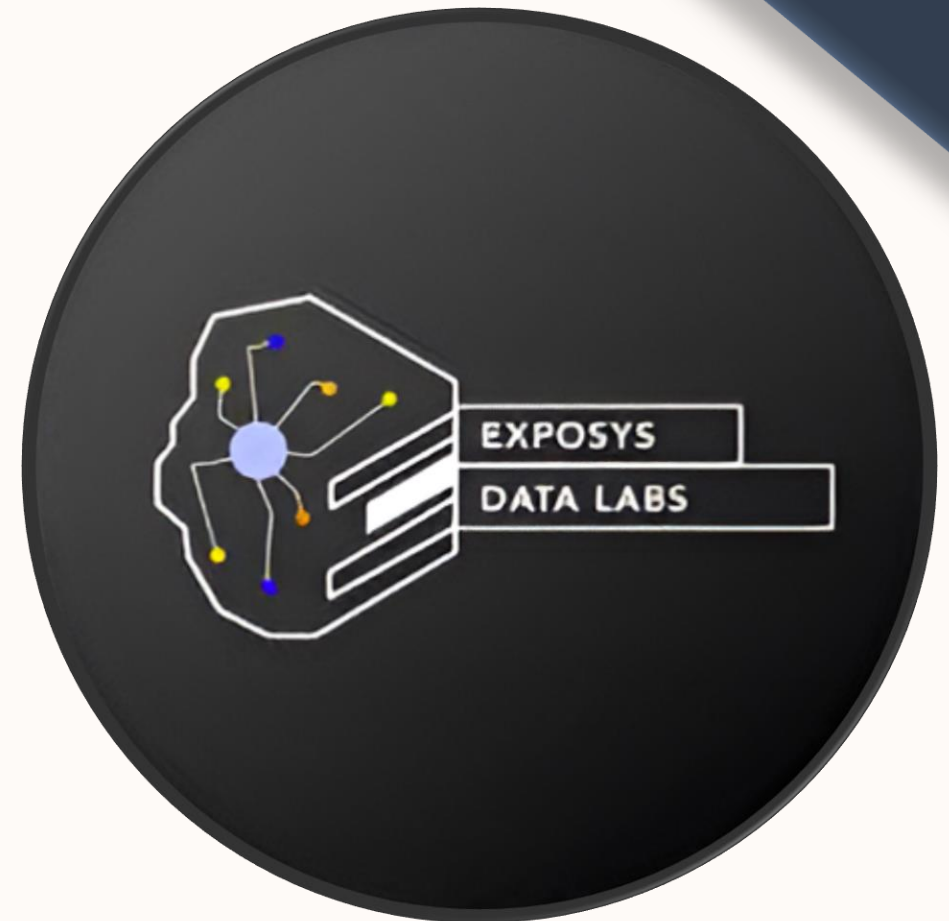


SECURED MESSAGE SENDING PLATFORM

Using Python and ML



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Abstract

Overview:

Development of a secure messaging application using Flask, Python, and advanced cryptographic techniques.

Key Features:

- Robust user authentication
- Encryption and decryption of messages
- Secure SQLite database for user management

• Goals:

- Enhance data security
- Maintain efficiency and usability

INTRODUCTION

Importance of Secure Communication:

- Growing digital communication demands high security
- Increasing risk of unauthorized access to personal messages

Project Objective:

- Develop a secure messaging platform with encryption and user authentication
- Ensure data integrity and confidentiality
- Provide a user-friendly experience

Existing Method

Current Issues in Messaging Systems:

- Weak password storage (plaintext or outdated hashing)
- Lack of end-to-end encryption
- Inadequate user authentication (no MFA)
- Vulnerability to message interception
- Poor session management

Proposed Method with Architecture

Encryption Techniques:

- Fernet symmetric encryption for secure message transmission

User Authentication:

- HMAC and SHA-256 for hashed passwords
- SQLite database for secure storage of user credentials and messages

Architecture:

- Flask web application for backend
- HTML and CSS for frontend interface

Methodology

Database Setup:

- SQLite database for managing user information and messages

User Registration and Authentication:

- Secure sign-up with hashed passwords

Message Handling:

- Encryption of messages using recipient's unique key

Web Development:

- Flask routes for user interactions
- HTML and CSS for user interface

Testing:

- Comprehensive testing for functionality and security

Implementation

Frontend:

- Clean, responsive user interface
- HTML forms for user input
- CSS for styling and usability

Backend:

- Flask framework for application logic
- Secure SQLite database management
- Secure user authentication and message encryption
- Flask routes and session management

Conclusion

Summary:

- Addressed need for secure communication with robust encryption and authentication
- Enhanced data security and communication integrity

Future Improvements:

- Integrate ML for anomaly detection in user behavior
- Explore additional encryption methodologies to combat evolving security threats

THANK YOU